

Social Cognition

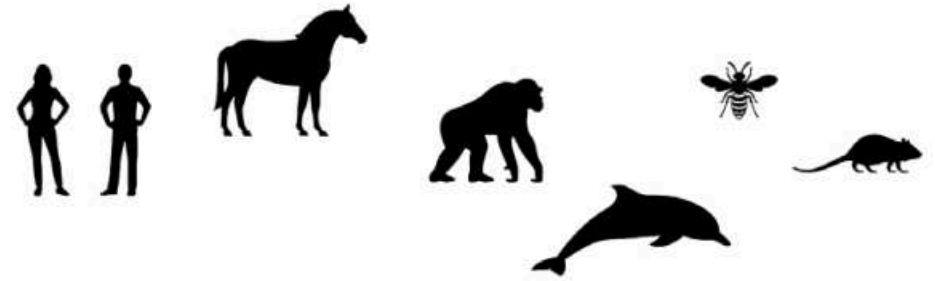
UE M1S1 - *Fundamentals in Neurosciences* - 2024

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Social cognition



- **Social cognition** refers to all the cognitive processes (perception, memorization, reasoning, emotions, etc.) that are involved in social interactions between **humans** and **social animal species**.
- Skills that enable an individual to (i) **recognize other individuals through a variety of means**; (ii) **recognize and remember relationships between other animals**; and (iii) **possibly attributing emotional and mental states to them**.

Social Learning

- Individual acquisition of new information in the context of social interactions with members of the same species.
- Fast way to learn new skills and knowledge about environmental events from experienced animals.
- Privileged channel for social cohesion and learning within a social group.



A widespread ability, from invertebrates to vertebrates



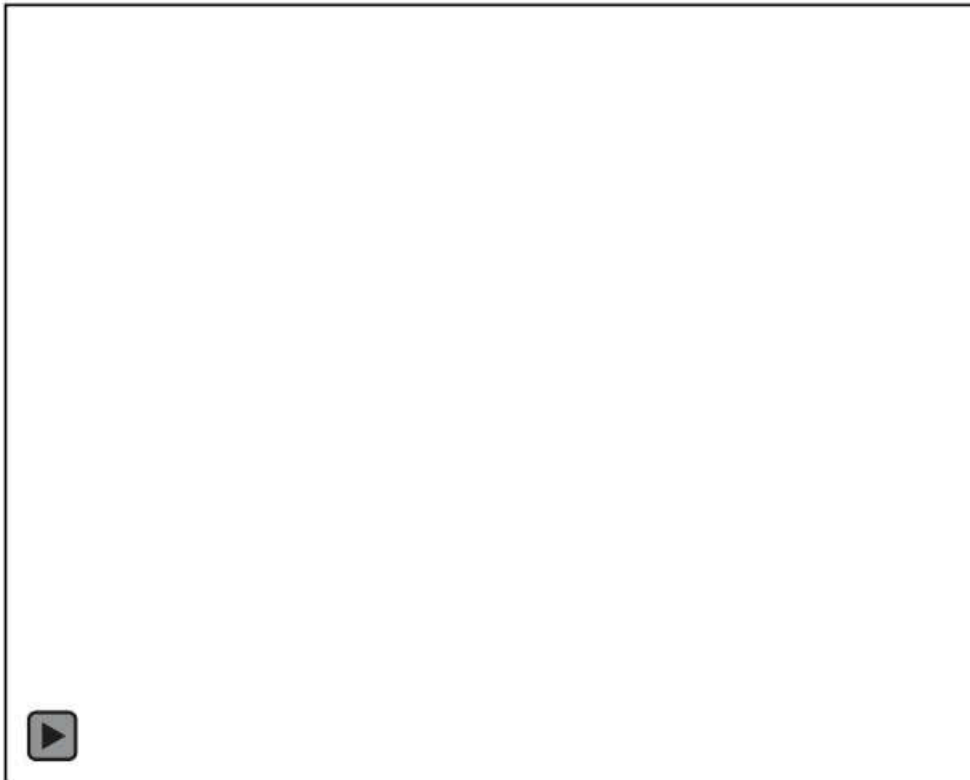
Japanese macaque washing a sweet potato on Koshima Island in Japan.

Potato washing among Japanese macaques on Koshima Island

- **Socially transmitted behavior**
- Transmission along a specific path. The spread of the behavior was between play partners and kinship relationships (children, mothers/younger siblings, older siblings)

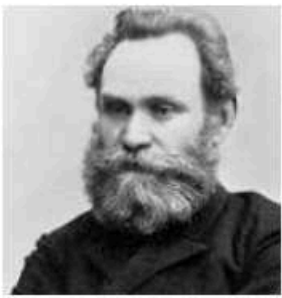
Invertebrates are no exception

Social Transmission of String-Pulling Test Resolution in Bumblebees

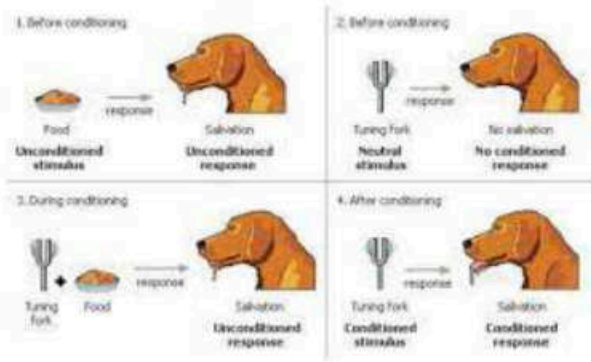


- The skill quickly spread from a **single innovative individual** to the majority of bumblebees in a colony.
- Naïve observers learned by interacting with innovative individuals and, subsequently, became demonstrators themselves for the next "generation" of learners.
- Thus, the competence in the population goes beyond the lifespan of the 1st innovators.

The representation of others: an associative hypothesis

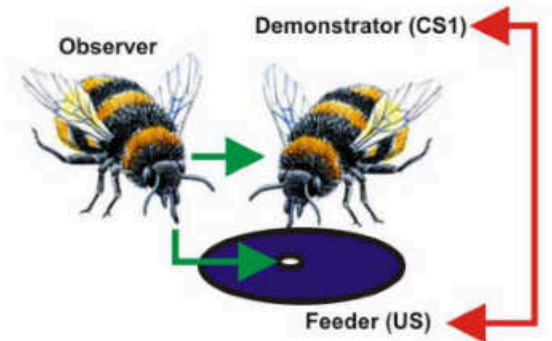


Ivan Pavlov
(1849-1936)



An Interpretation based on **2nd Order Pavlovian Conditioning**

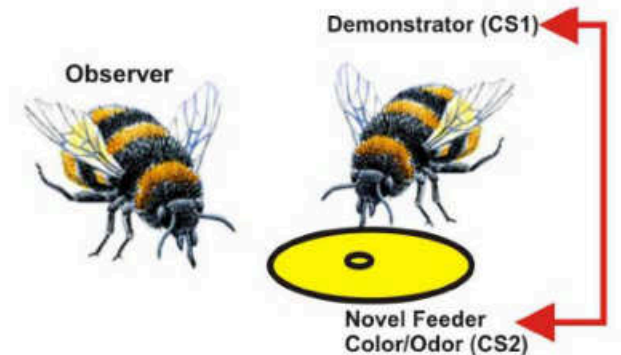
Pavlovian Conditioning: Acquisition of an Association between a **Conditioned Stimulus (CS)** and **Unconditioned Stimulus (US)**



2nd order Pavlovian conditioning: Acquisition of 2 consecutive "connected" associations

CS1 → US

CS2 → CS1 (thus US)

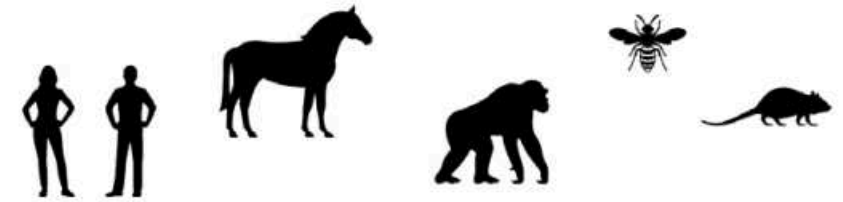


Social cognition goes beyond simple associative learning and can engage complex cognitive processes

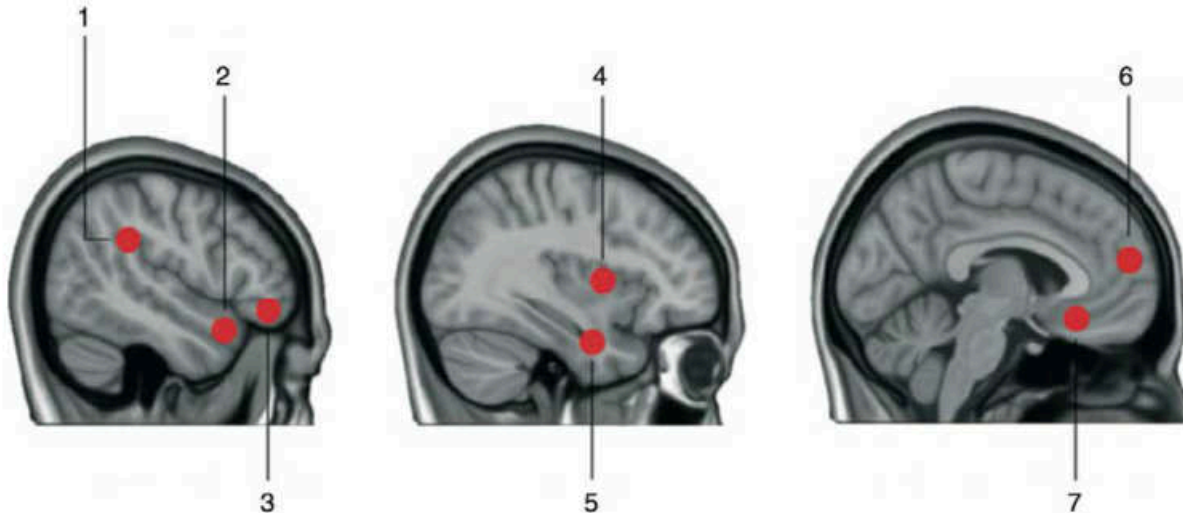
More complex representations can be involved, particularly in tasks where **empathy** intervenes: **the ability to represent and understand the emotional and cognitive state of others.**



Neuroanatomical correlates of theory of mind



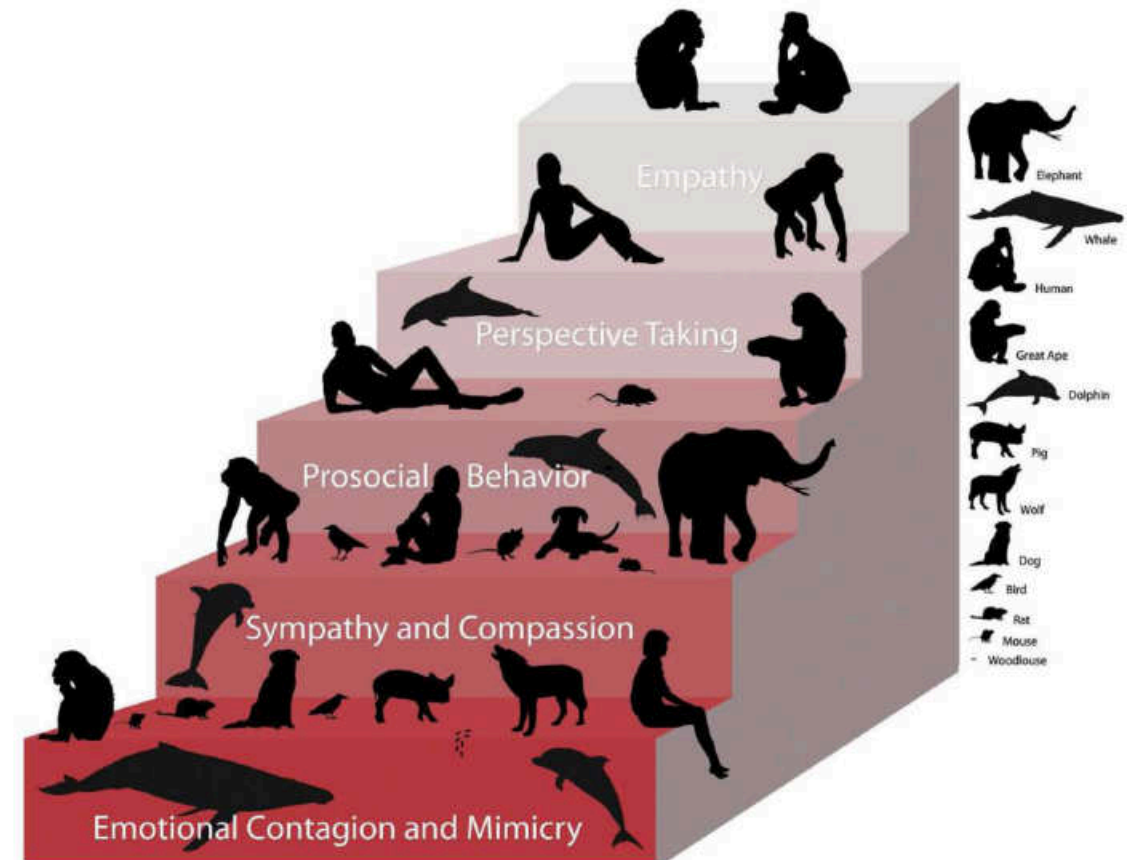
EMC



- 1- Jonction temporo-pariétale
- 2- Pôle pariétal
- 3- Cortex orbito-frontal
- 4- Insula
- 5- Amygdale
- 6- Cortex préfrontal médian
- 7- Cortex préfrontal ventromédian

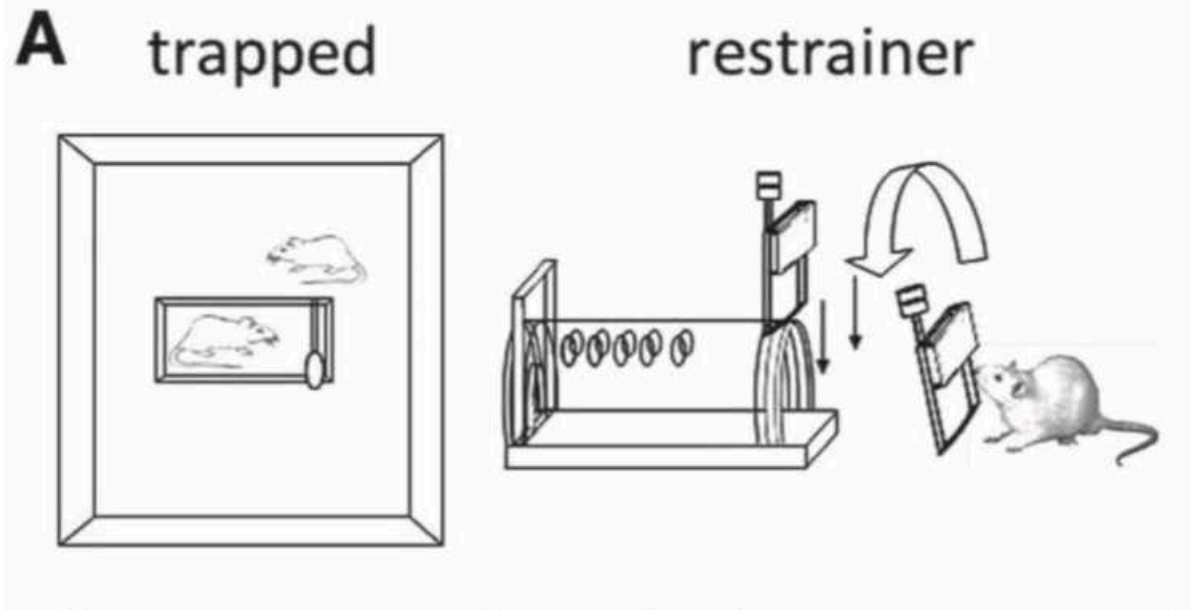
Social Cognition: Empathy and Related Behaviors in Animal Models

- Several empathetic behaviors have been documented in many animal species.
- Different levels are recognized in these behaviors until they reach the true level of empathy.
- **Copying the behavior of others, preferential affiliation, prosocial behavior (benefiting someone even with personal costs), change of perspective; Emotional and cognitive empathy.**



Empathy and pro-social behavior in rats

A free rat in an arena where another rat is locked within a container (stressful situation)

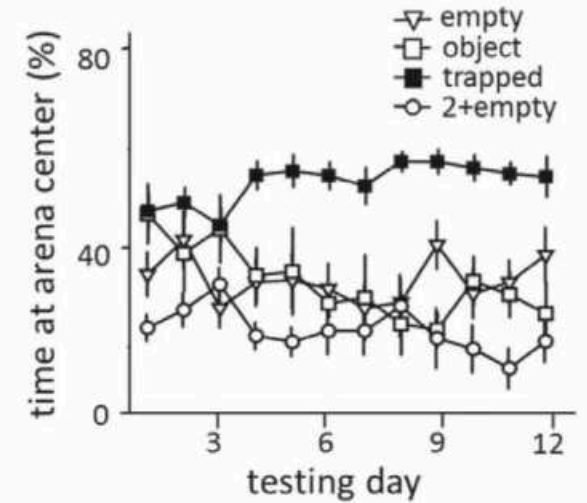
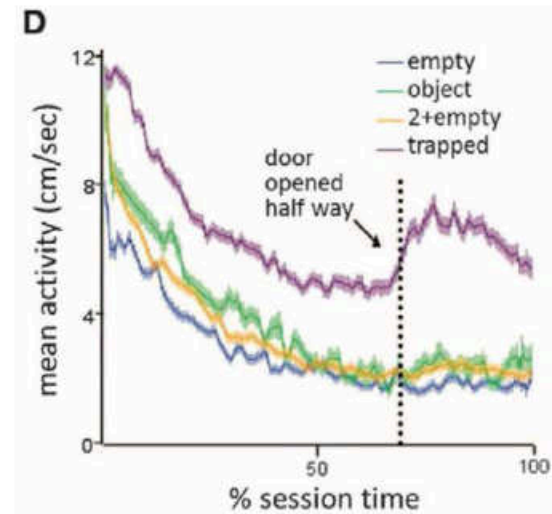
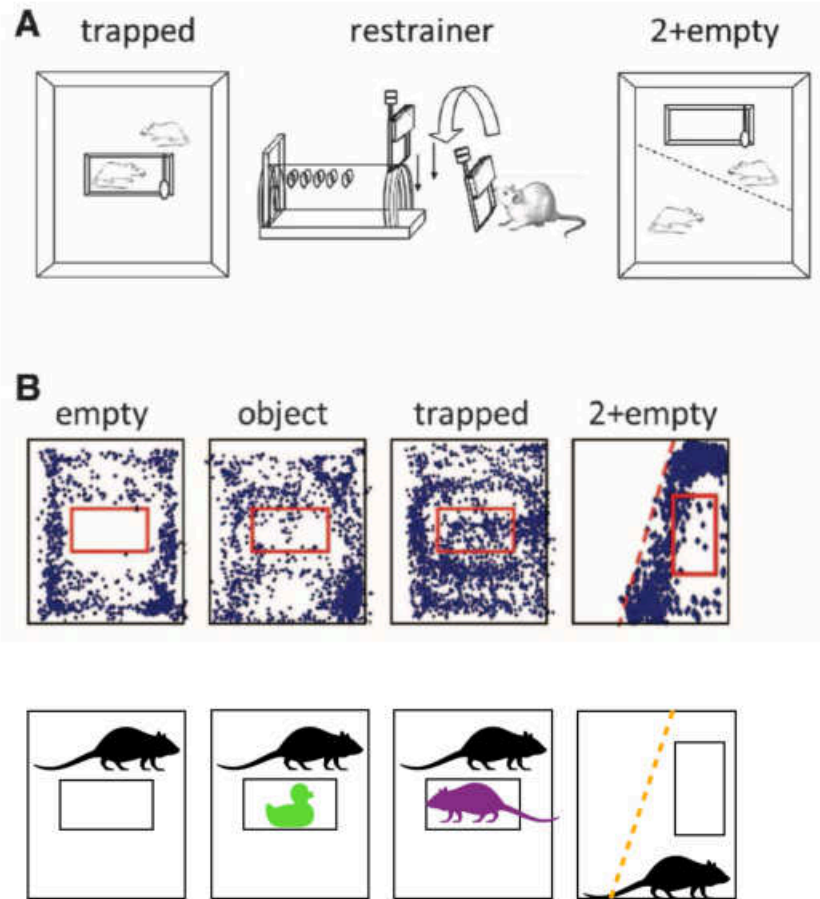


A rat learning to open the restrainer

across 12 days of testing,
A rat is placed with a trapped cagemate.
The rat does not know how to open the door on day 1.
On day 5 he opens the door for the first time.
By day 12, the rat is an expert opener.

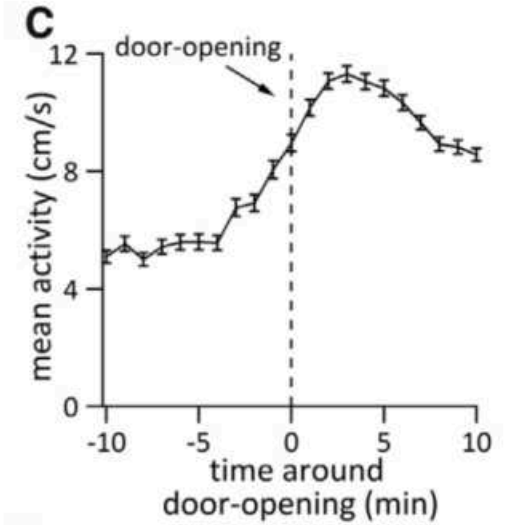
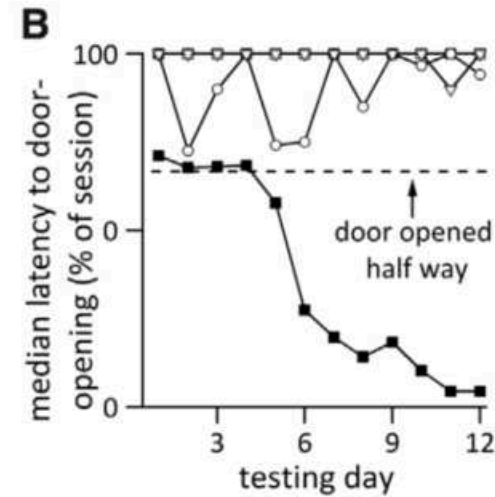
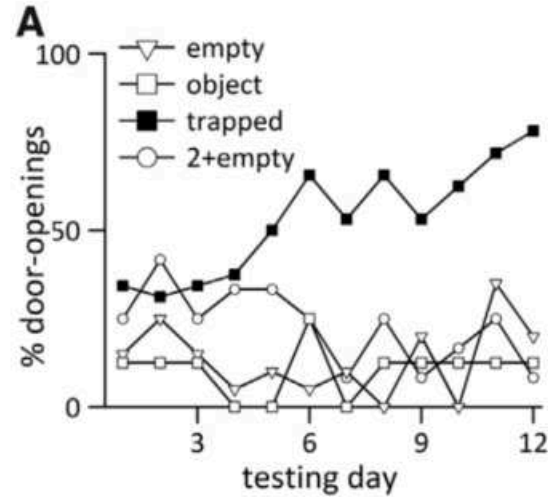
- After several sessions, the free rat learns to intentionally and quickly open the container and release the trapped rat.
- The rats did not open empty containers or containers containing objects.

Empathy and pro-social behavior in rats



Rats show greater locomotor and exploration activity when their congener is trapped

Empathy and pro-social behavior in rats



Only rats in contact with their congener trapped show:

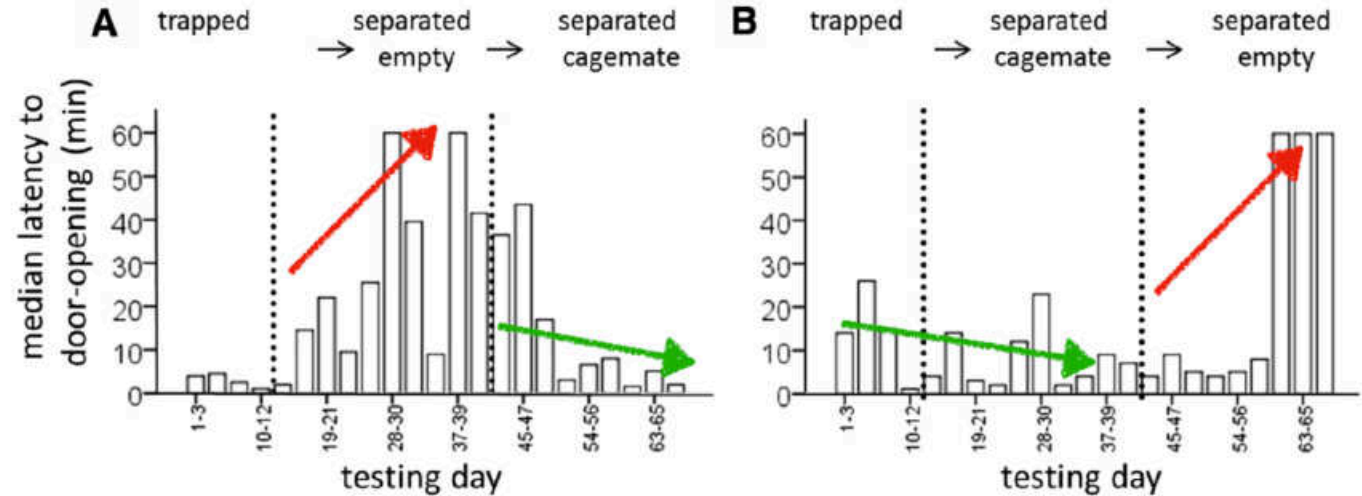
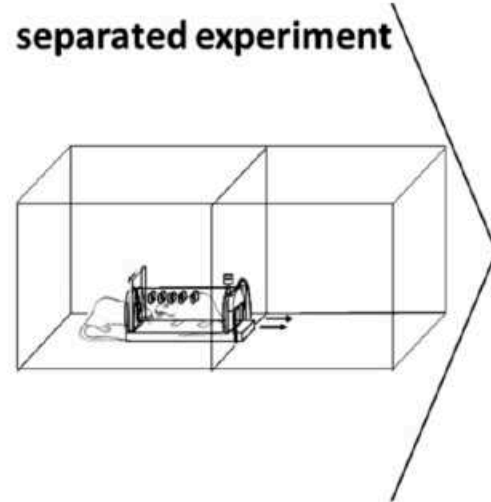
- an increase in the opening of the trap during the tests
- A decrease in the latency to open the trap.
- An increase in activity when the trap has been opened (time 0 in C)

Rats adapt their behavior only in the presence of their congener trapped

Empathy and pro-social behavior in rats

Is it empathy or the search for social contact as a “personal” reward?

Opening the hatch allows exit to another arena without social contact



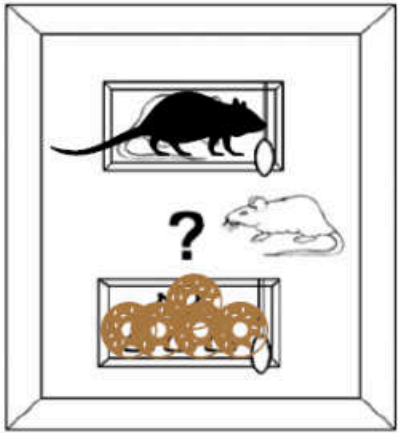
The reduction in the latency of opening the hatch depends on the presence of a conspecific, regardless of social contact

Rats release their captive fellow despite the lack of subsequent social interaction.

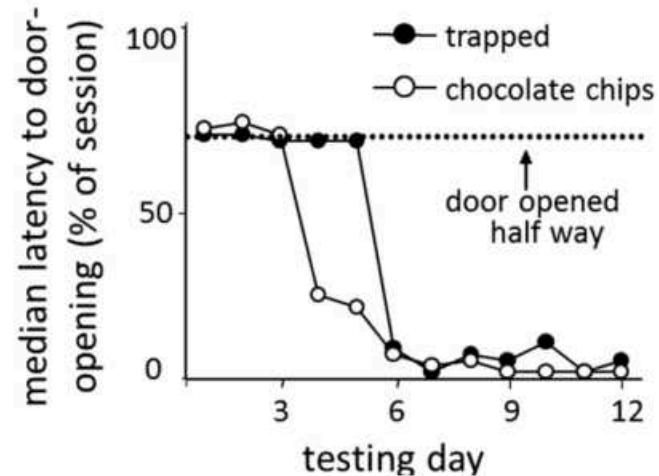
Empathy and pro-social behavior in rats

Priority to the empathetic act or to the individual reward?

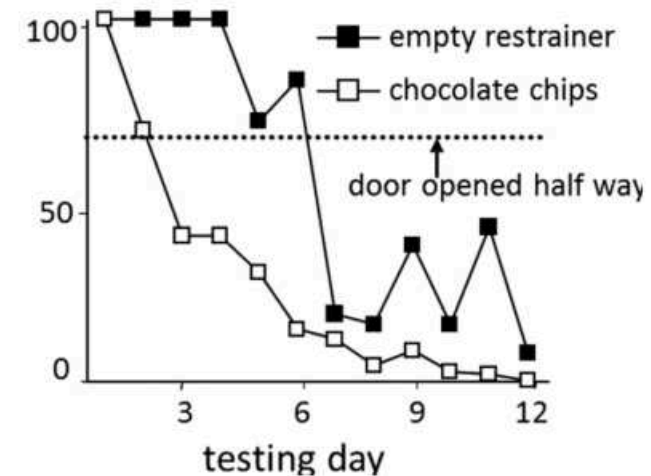
Paradigm of **congener enclosed vs. chocolate**



C chocolate cagemate condition



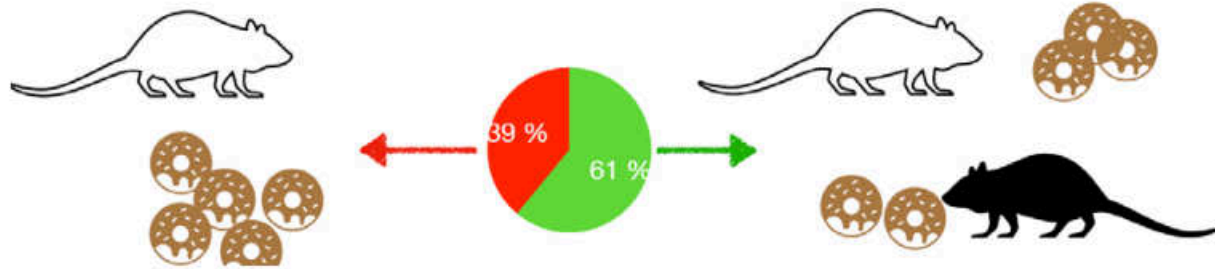
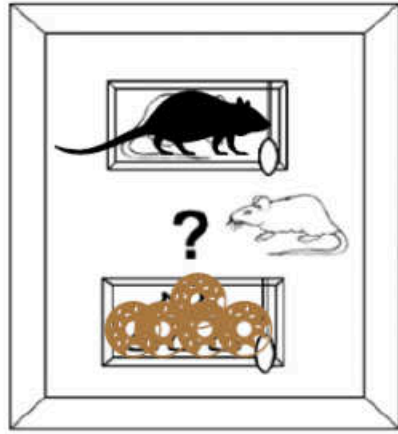
D chocolate empty condition



- The latencies to open the hatch in days 6-12 are **similar between chocolate and congener conditions**
- The latencies to open the hatch are longer for the empty container compared to those where there is chocolate

The value of the empathic act is similar to that of the individual food reward

Empathy and pro-social behavior in rats



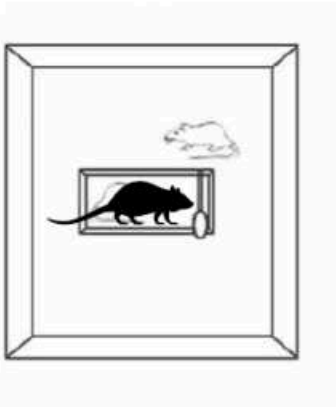
Empathetic behavior leads to food sharing

Empathy and pro-social behavior in rats

- Motivation to open the door and free the congener:
- Contextual clue unrelated to the distress of the captive animal.
- Stop a distress signal (stress pheromone/alarm vocalization...) → Affective transference
- The more plausible explanation: Rats release their congener in order to end the distress, their own or that of the trapped rat, which is associated with the circumstances of the trapped cage mate.
→ Empathy

Emotional state is a key factor in expressing pro-social behavior

Role of emotional state

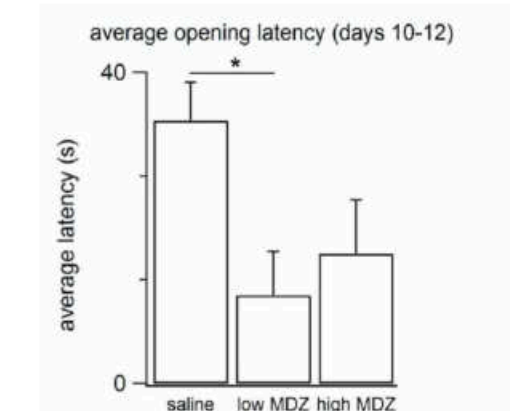
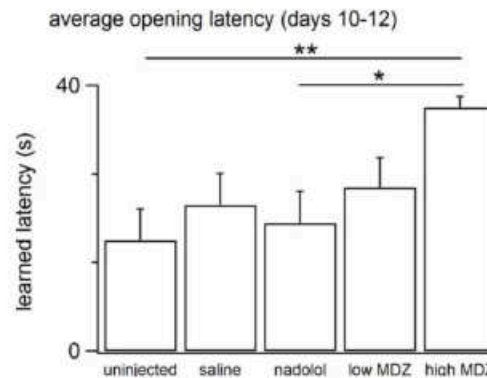
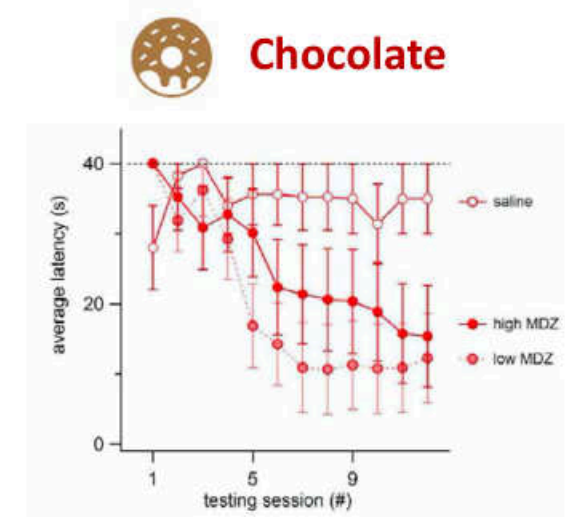
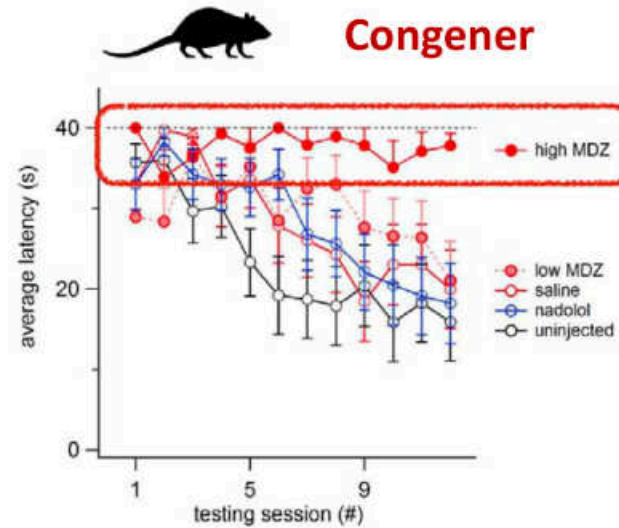


MDZ midazolam : benzodiazepine (anxiolytic)
Free rat injection:
reduction of individual stress

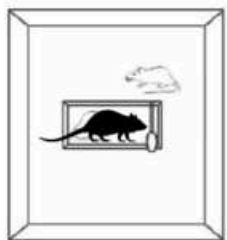
Nadolol : Beta-adrenergic peripheral
(sympathetic) blocker

Rats treated with midazolam showed less helpful behavior.
Midazolam has no effect in the case of a container containing chocolate, thus showing the socially specific effects of the anxiolytic.

Ben-Ami Bartal & al, 2016 Frontiers in Psychology 7:850



Emotional state is a key factor in expressing pro-social behavior

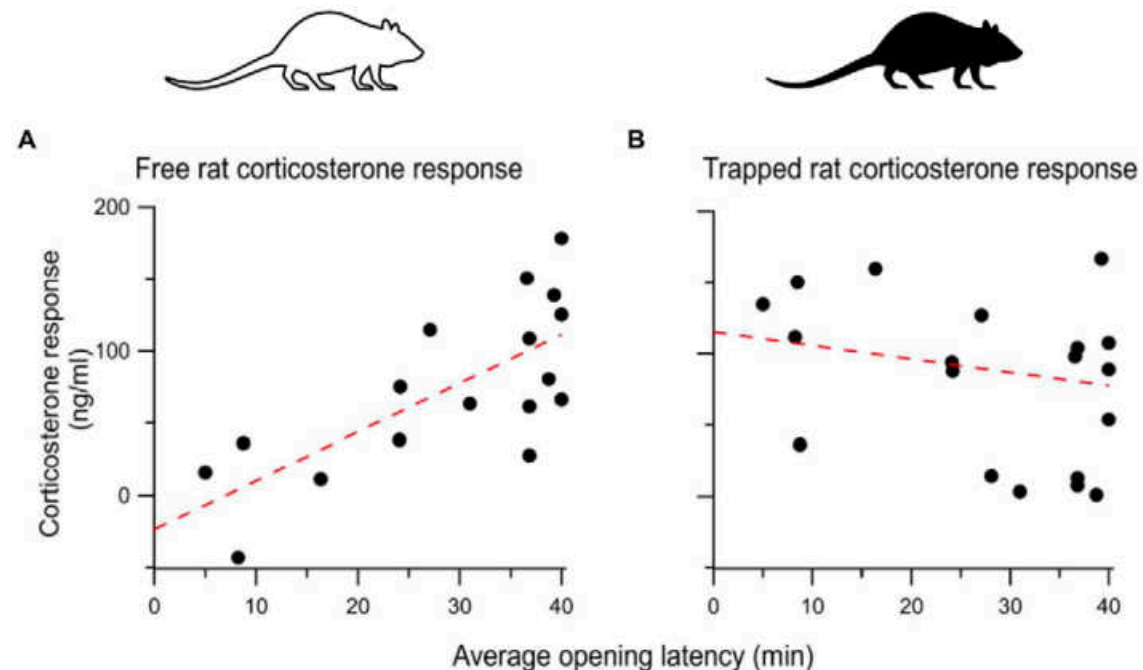


Door blocked to impose different latencies on the free rat

Measure of HPA axis (hypothalamic–pituitary–adrenal axis)
/ **Corticosterone level (stress level)**

Positive correlation between corticosterone level and mean hatch opening latency only in free rats.

No significant correlation in enclosed rats.

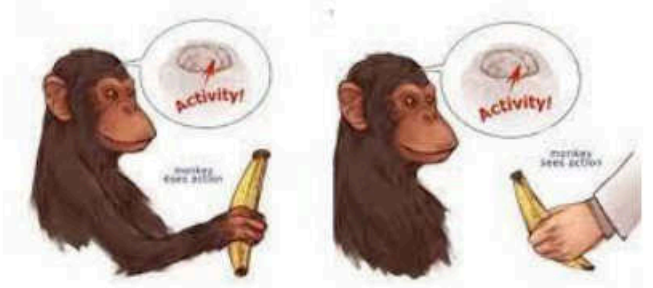


The inability to open increases the stress of the free rat.

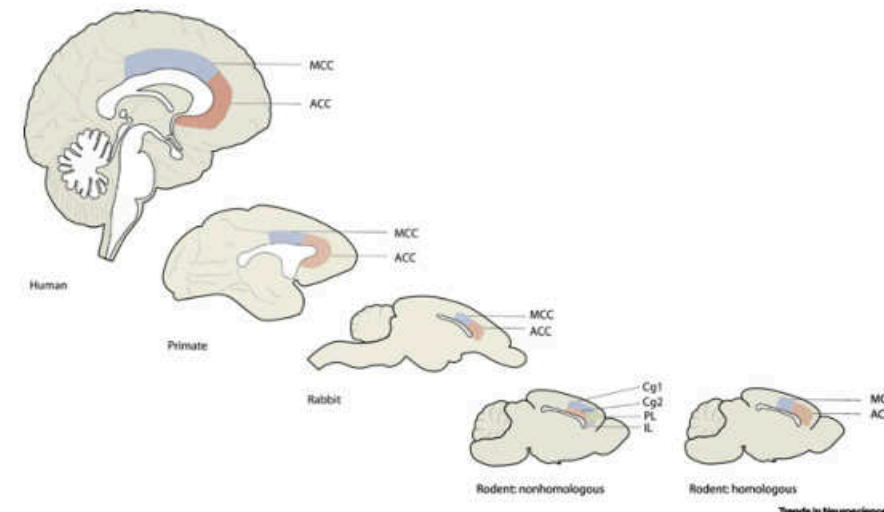
The emotional state of the free rat is decisive for empathic behavior.

The neural bases of emotional contagion: emotional mirror neurons in the ACC (anterior cingulate cortex)

- Does the rat's ACC contain **mirror neurons for pain**, i.e., neurons that respond to the own pain experience and to another individual's pain perception?
- Are these neurons **emotion-specific**? Do they respond differently to other emotions distinct from pain? (**Selectivity**)
- Do these neurons use a **common code** to represent the own pain and the pain observed in others?
- Is the **ACC necessary for emotional contagion**? (deactivation by Muscimol injection: GABAA agonist receptor)

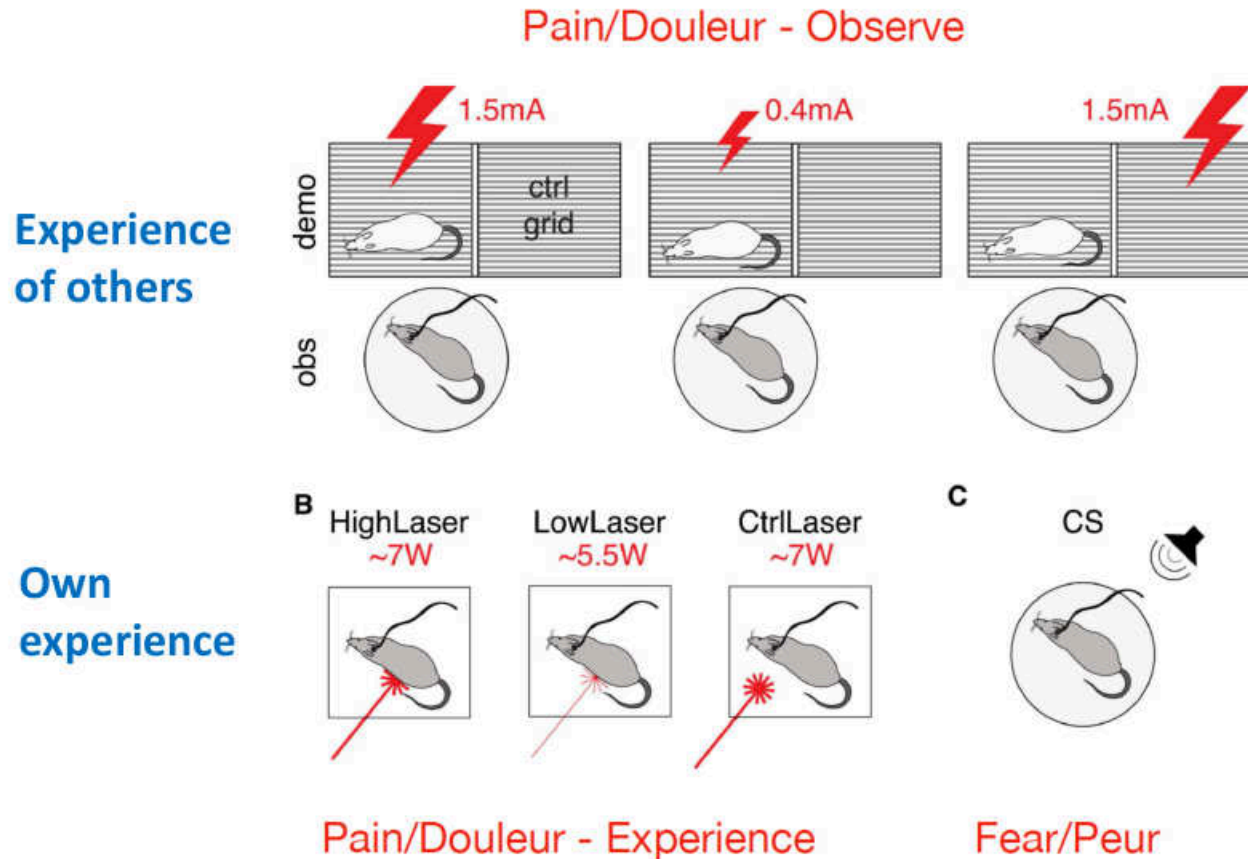


Mirror neurons respond when an individual performs an action and when it observes another individual of the same species performing the same action or when it imagines the action being performed



Emotional contagion protocol

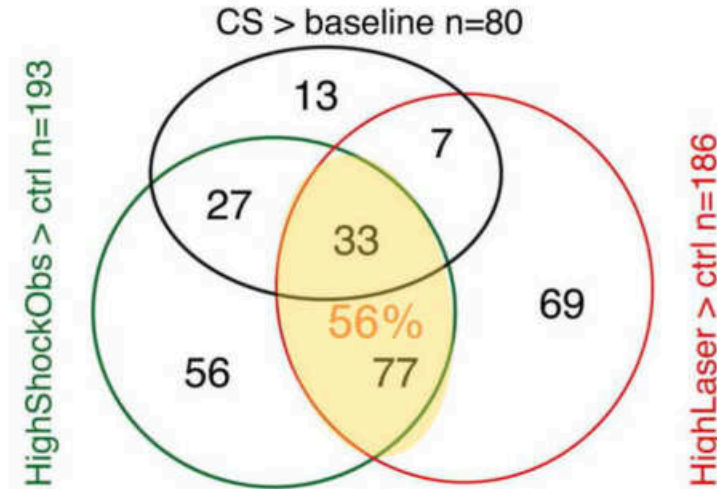
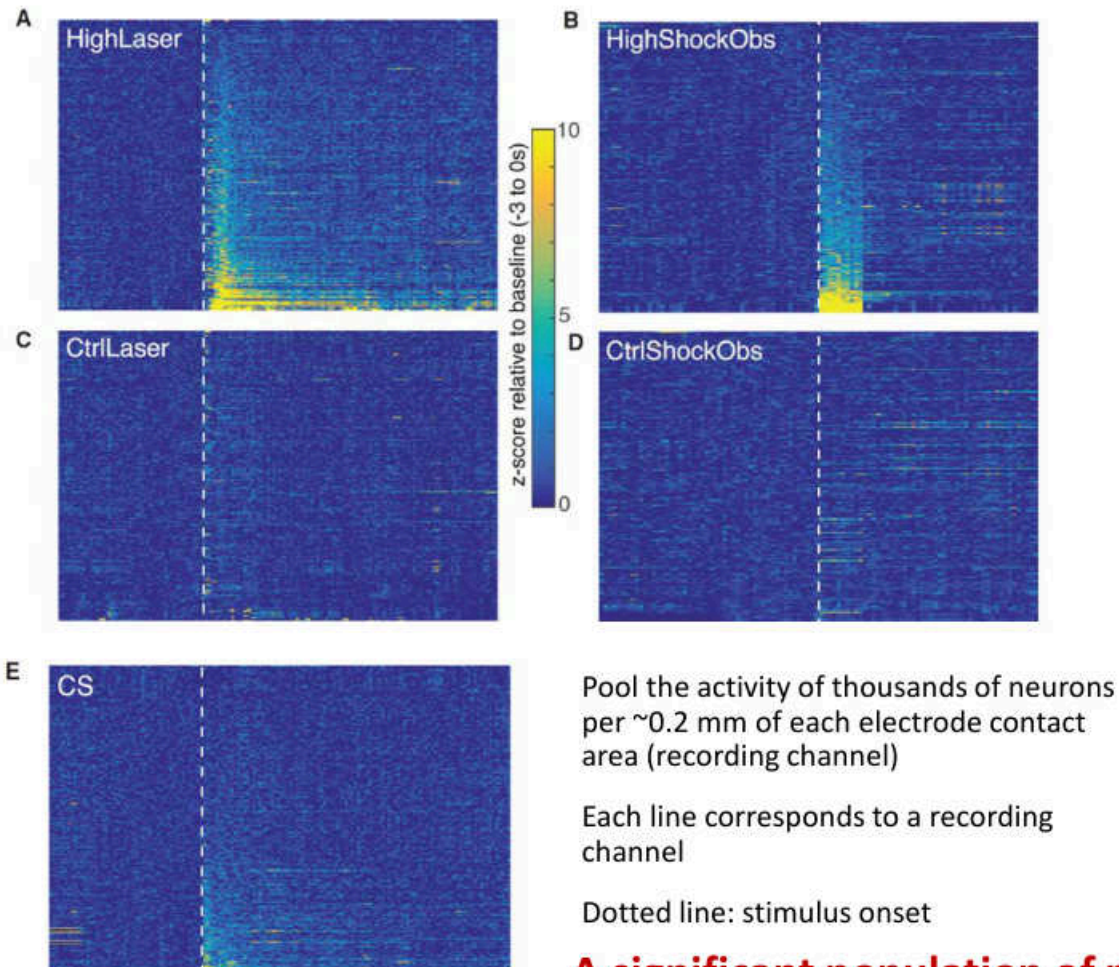
Observer rats **pre-exposed to shock**, before a test with demonstrator rats subjected to high-level and low-level shocks and to the CS (tone)



Two different emotions tested: pain (shock/laser) vs. Fear (CS)

Multi-unit records in the observer rat ACC

Venn diagram for number of channels



In the ACC, a population of neurons

- Responds to individual pain (Laser)
- Responds to the expression of pain in a congener (vocalization)
- Responds to both stimuli

A significant population of neurons responds to both pain and the observation of pain in a congener

Recording of single neurons in the ACC

Responses of individual neurons isolated by spike sorting

In ACC, individual neurons react:

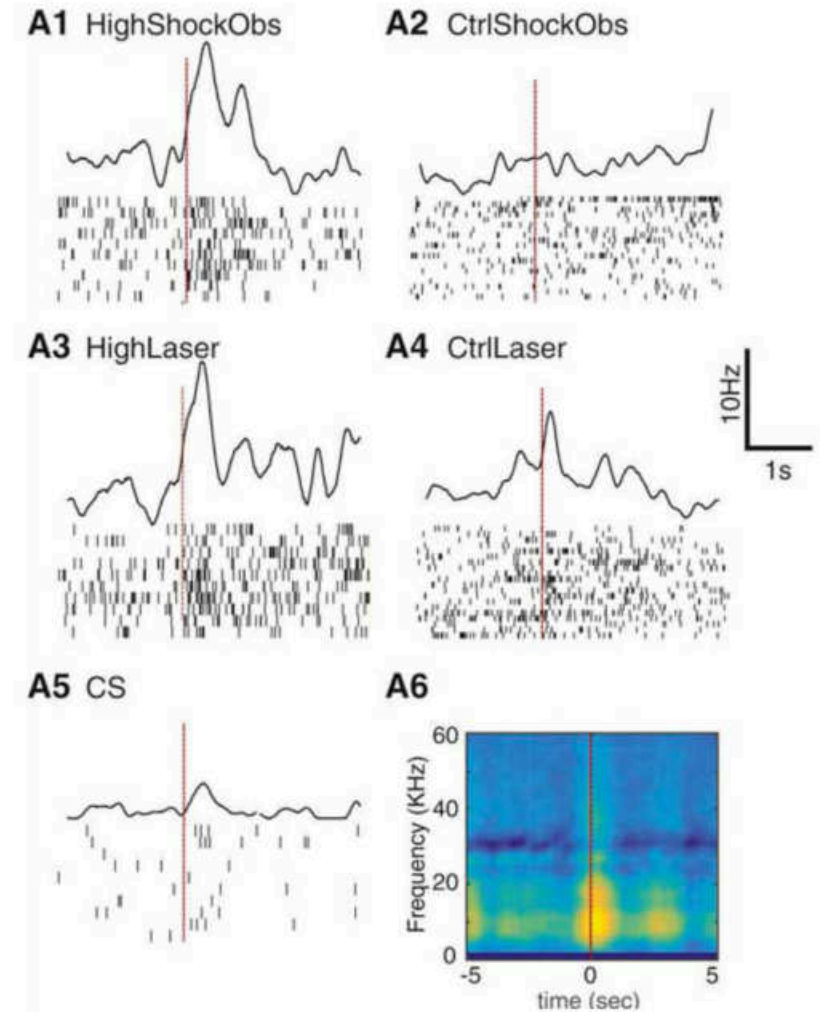
- to the pain observed in the congener (vocalizations).
- to the pain felt (Laser).
- to both stimuli.

Selective emotional neurons in the ACC, i.e., neurons responding to pain observation (shock) also respond to the actual experience of pain (laser) but not to other non-painful salient stimuli (CS --> fear).

Pain observed in congeners

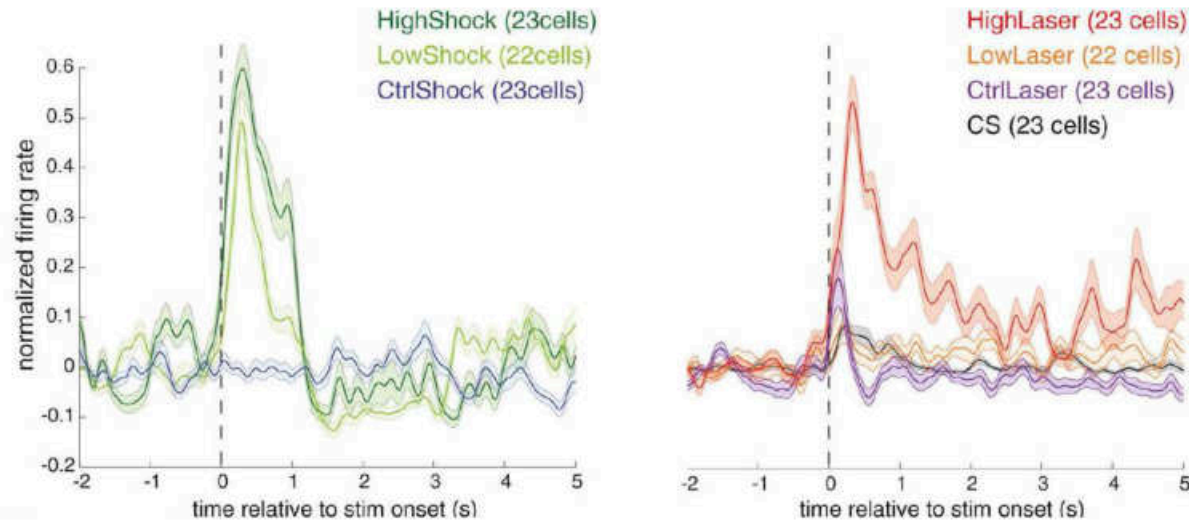
Own experienced pain

Own experienced fear



→ The ACC contains neurons that map another animal's pain response to a shock on representations of own pain.

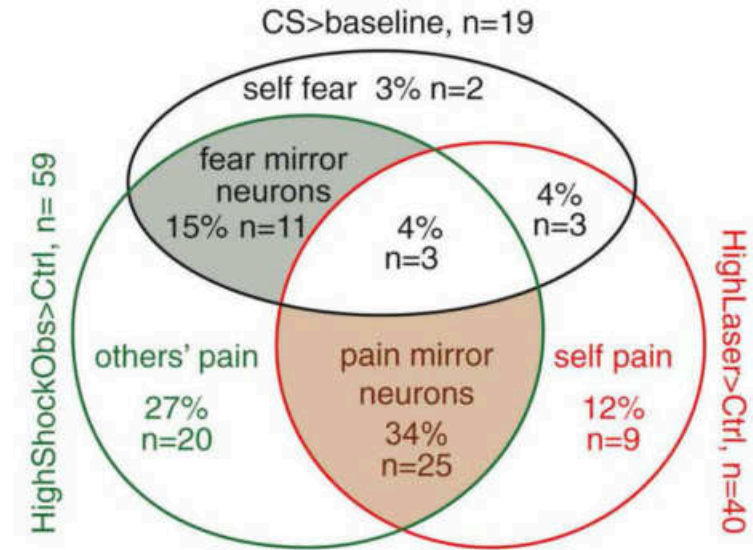
Recording of single neurons in the ACC



Response profiles of pain-selective mirror neurons

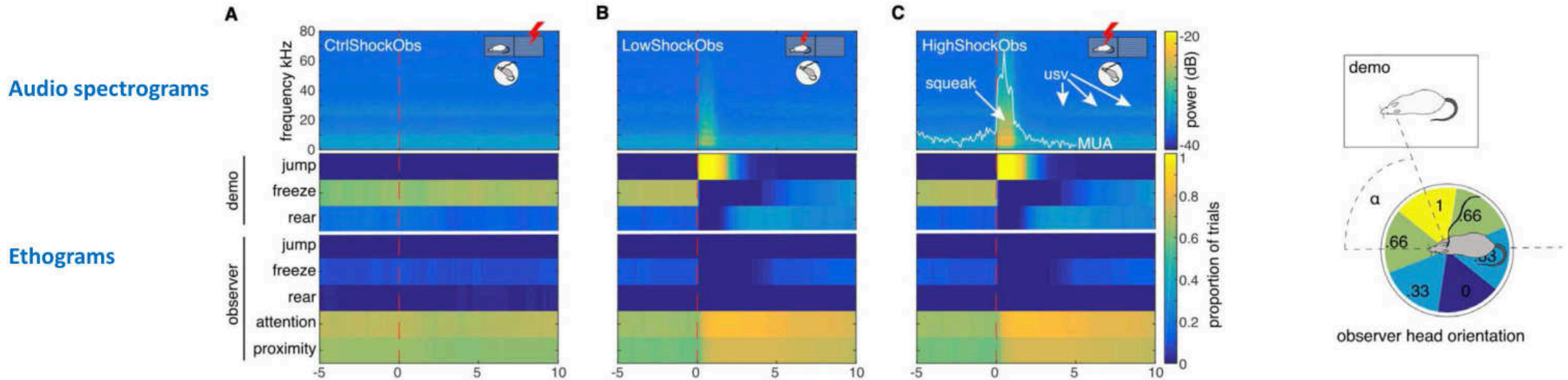
25 neurons identified as pain mirror neurons:

- Strong response to pain observation (high and low intensity shock)
- Strong response to high-pain experience only (laser: own pain)



Demonstration of Pain Mirror Neurons in the rat ACC

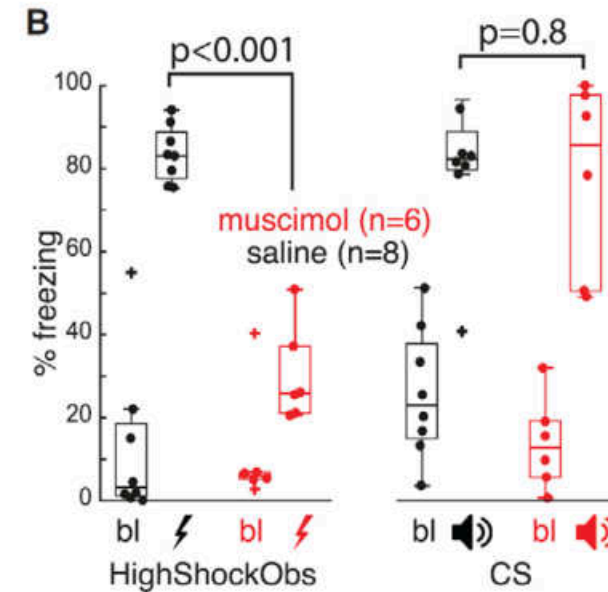
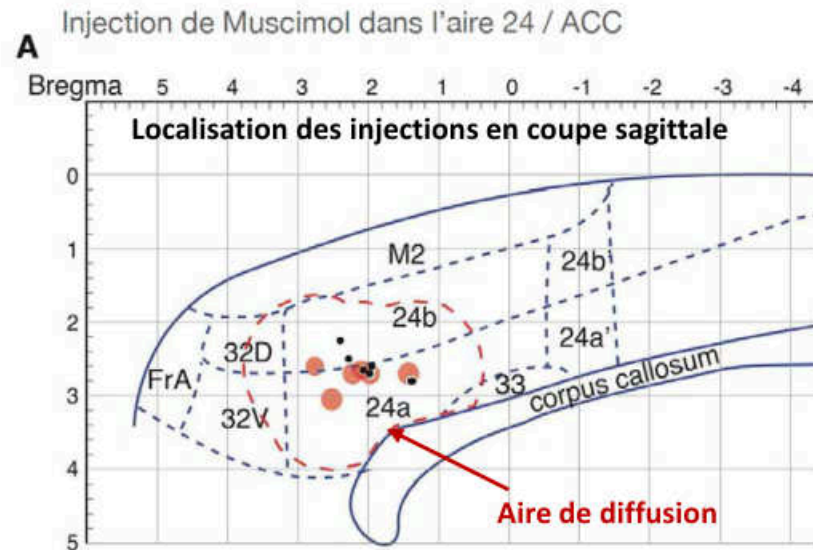
Behavioral quantification of responses during electric shock experiments



- The shock leads to a systematic "squeak" vocalization in the **demonstrator**, correlated with the response of the observer's neural ACC populations (HighShockObs condition).
- The shock leads to specific behaviors in the **demonstrator** (jump).
- These **demonstrator behaviors** are associated with the **observer's closeness and orientation** towards the demonstrator (attention and proximity)

The ACC is necessary for Emotional Contagion Behaviors

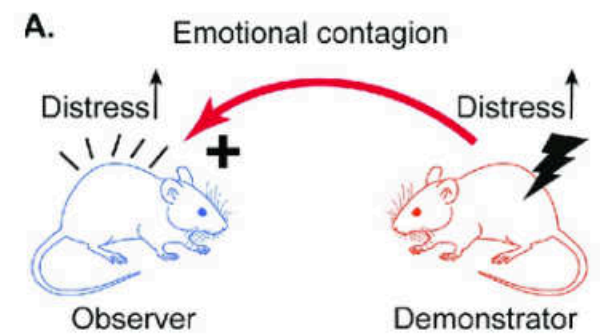
ACC deactivation by localized injection of muscimol



Freezing triggered socially by observation of the shock of others was reduced in the group injected with muscimol compared to the group injected with saline. This difference does not appear in the non-social condition (freezing in CS)

ACC is necessary for the expression of aversive behavior (freezing) only under conditions of social contagion

Neural bases of social contagion: some conclusions



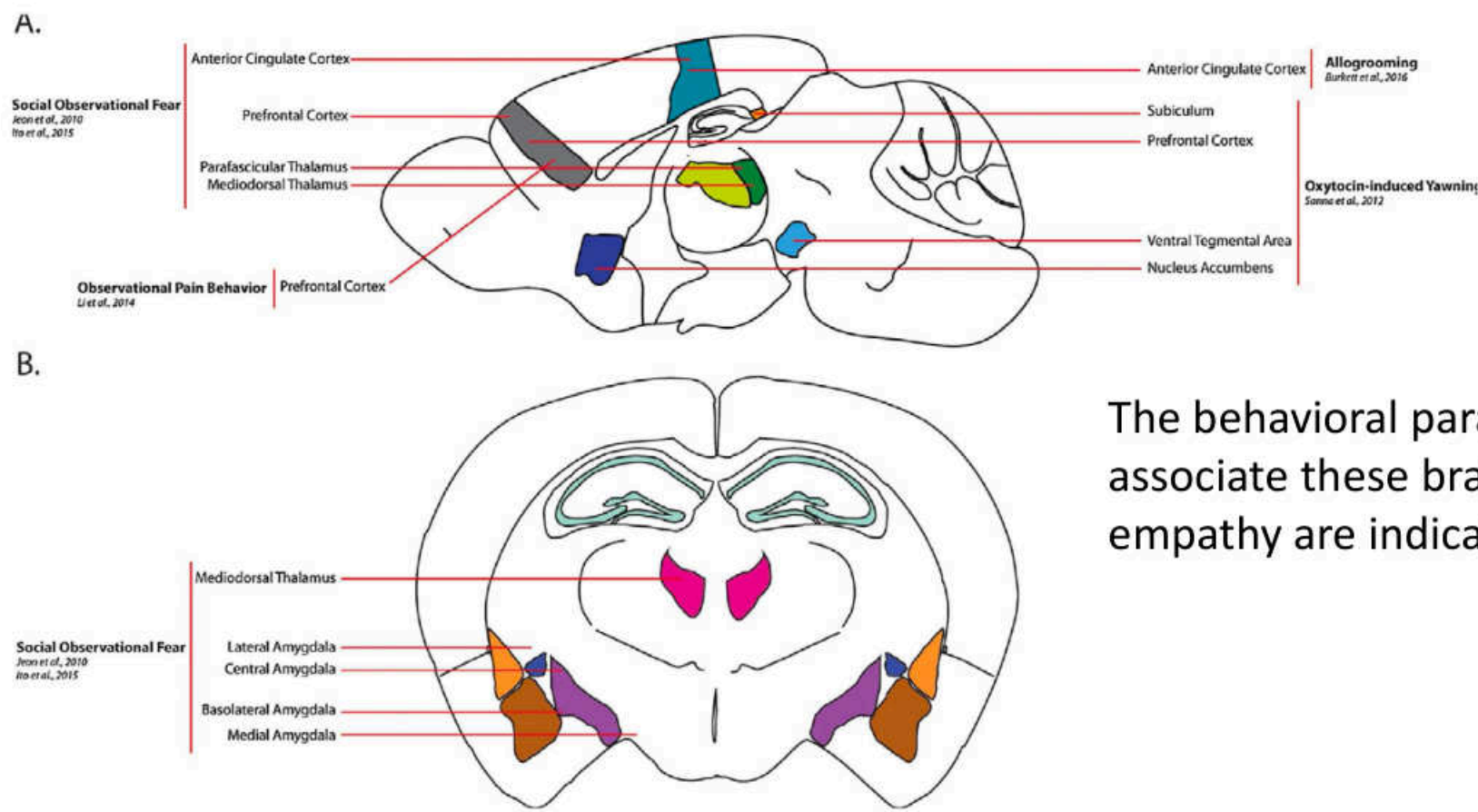
- The rat ACC shows neuronal activity, at the level of populations of neurons or individual neurons, of the selective mirror neuron type during the observation of shocks in another rat and during the own experience of pain (Laser).
- This selectivity is demonstrated by the lack of neural activity in these neurons in the case of another emotion, fear at a sound CS.
- Deactivation of the ACC reduces socially triggered freezing without compromising freezing to non-social danger signals (CS).
- Pre-exposure to the shock in the observer determines his or her emotional contagion. What would happen in the case of laser pre-exposure?

WARNING: emotional contagion does not necessarily imply empathy!

- It can prepare the observer to deal with the pain affecting the demonstrator.
- But it is not enough to show that the observer understands that this pain is felt by another animal.



Neural Basis of Social Cognition: A Summary of Several Studies on Empathy in Rodents



The behavioral paradigms used to associate these brain regions with empathy are indicated

Social cognition in animals

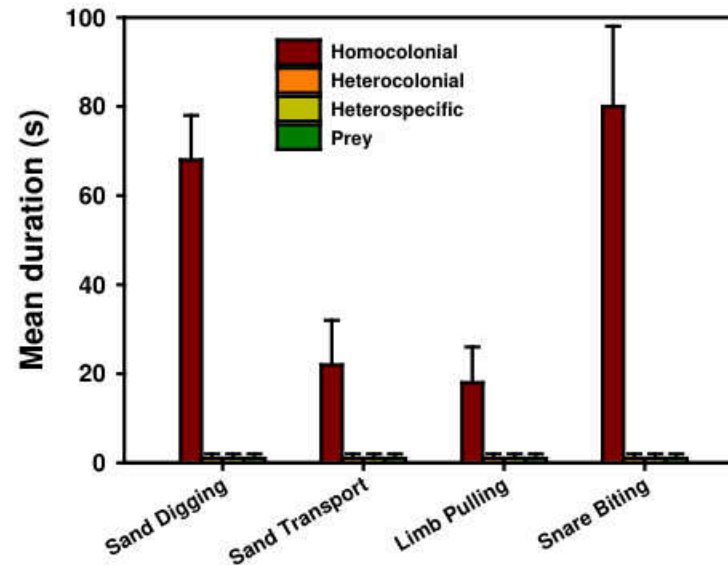
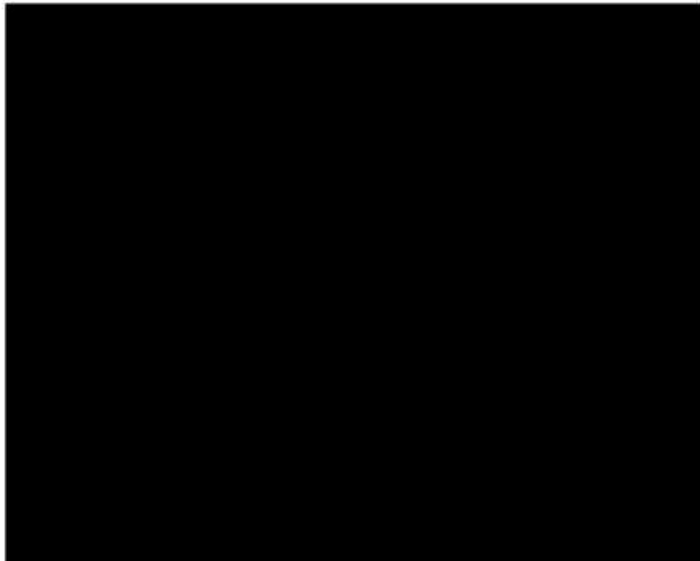


- Social cognition: functions and identification in animals
- Empathy? Theory of mind?
- Reverse translational research: from humans to non-human animals
- Neurobiological correlate
- Cellular and molecular bases
- Identification of disorders in pathologies of social cognition (ASD, Schizophrenia, neurodegenerative diseases, etc.)



Finally... do not "vertebrize"...

Empathic-like behavior in ants



Cataglyphis cursor

- *Cataglyphis cursor* ants engage in a specific behavior to release conspecifics attached by a nylon thread at the level of the leg to a hidden cardboard, buried in sand.

- The ants excavate the sand, exposing the nylon thread, carry the sand away from it, and then bite the thread itself to free the trapped ant.
- They distinguish between ants in distress, **offering help only to ants of the same species and of the same nest.**

CONTROLS NEEDED

- Pheromones: reinforcement of operant behaviors?
- Experiments in which objective devaluation is performed must also be carried out. Causal sensitivity must be proven.